

SuSh at MultiClinAI 2026: Zero-Shot Multilingual Clinical Named Entity Recognition Using GLiNER-biomed-large

Anonymous ACL submission

Abstract

We present the SuSh system for Subtask 1 of the MultiClinAI shared task at the 11th SMM4H and HearD Workshop (ACL 2026), which addresses multilingual clinical named entity recognition (NER) across seven languages. Our system adopts a fully zero-shot approach using GLiNER-biomed-large-v1.0, a span-based NER model pre-trained on biomedical text, requiring no task-specific fine-tuning or labeled data in target languages. We apply a character-level sliding window strategy to handle long clinical documents that exceed the model’s token limit, and incorporate a post-processing pipeline including threshold optimization via F1-max sweep, entity-specific gazetteer lookup derived from DisTEMIST and SympTEMIST terminology lists, span boundary correction, and negation filtering. Our official submission achieves a Strict F1 of **0.5175**, Strict Precision of **0.5536**, Strict Recall of **0.4859**, and CHR F1 of **0.6130** on the English disease subtask, demonstrating that domain-adapted zero-shot biomedical NER models can serve as competitive baselines for multilingual clinical entity recognition without any task-specific training data.

1 Introduction, Research Question, and Objectives

Clinical named entity recognition (NER) is a foundational task in biomedical natural language processing, enabling downstream applications such as clinical decision support, pharmacovigilance, cohort identification, and electronic health record mining. The MultiClinAI shared task (Gallego-Donoso et al., 2026; Lopez-Garcia et al., 2026) at the 11th SMM4H and HearD Workshop challenges participating systems to identify three clinical entity types (diseases, symptoms, and medical procedures) across clinical text in seven languages: English, Spanish, Czech, Dutch, Italian, Romanian, and Swedish. The task is organized by

the Barcelona Supercomputing Center (BSC) as part of the DataTools4Heart and AI4HF European projects.

Research Question. *Can a domain-adapted zero-shot biomedical NER model, augmented with targeted post-processing, achieve competitive performance on multilingual clinical NER without any task-specific fine-tuning or labeled training data?*

Objectives. Our specific objectives are: (1) to evaluate the zero-shot transfer capability of GLiNER-biomed-large-v1.0 on the Multi-ClinAI English disease subtask; (2) to design a post-processing pipeline that improves Strict F1 over the raw model baseline through threshold optimization, gazetteer augmentation, span boundary correction, and negation filtering; and (3) to analyze error patterns to understand the limitations of zero-shot clinical NER and motivate future supervised approaches.

2 Background and Related Work

2.1 Clinical Named Entity Recognition

Clinical NER has progressed from early rule-based systems and dictionary-based approaches through statistical sequence labelers (CRF, HMM) to deep learning models based on bidirectional LSTMs and transformer architectures (Li et al., 2020). A persistent challenge is the scarcity of labeled clinical corpora, which are expensive to annotate due to privacy constraints and the requirement for medical expertise.

2.2 Transformer-Based Biomedical NER

BERT-based models (Devlin et al., 2019) and their biomedical variants such as BioBERT (Lee et al., 2020) and BioClinicalBERT (Alsentzer et al., 2019) have become standard baselines for clinical NER, typically fine-tuned using IOB2 token

classification. For multilingual settings, XLM-RoBERTa (Conneau et al., 2020) provides strong cross-lingual transfer through pretraining on 100 languages, enabling zero-shot or few-shot transfer to low-resource target languages.

2.3 Span-Based and Zero-Shot NER

A limitation of token classification models is their dependence on fixed task-specific label sets. GLiNER (Zaratiana et al., 2024) addresses this by framing NER as a span-entity matching problem: given descriptive entity type labels at inference time, the model scores all candidate spans against each label using a shared bidirectional encoder, enabling zero-shot generalization without retraining. GLiNER-biomed (Yazdani et al., 2025) extends this framework with biomedical domain adaptation, training on curated biomedical NER datasets to improve performance on clinical and scientific text.

2.4 Multilingual Clinical NER Resources

The MultiClinAI task builds on a series of BSC shared tasks, including DisTEMIST (Miranda-Escalada et al., 2022) for disease detection and SympTEMIST (Lima-López et al., 2023) for symptom recognition in Spanish clinical text. Lima-López et al. (Lima-López et al., 2024) explore annotation projection via neural machine translation as a strategy for generating multilingual clinical NER corpora from Spanish gold-standard annotations, motivating the MultiClinAI Subtask 2 annotation projection challenge.

3 Proposed Approach

The SuSh system follows a zero-shot inference pipeline consisting of document chunking, model inference, and a four-stage post-processing pipeline.

3.1 Base Model: GLiNER-biomed-large

We use `Ihor/gliner-biomed-large-v1.0` (Yazdani et al., 2025), a variant of the GLiNER architecture adapted for the biomedical domain. The model takes both document text and descriptive entity type labels as input, scoring all candidate character spans against each label. This enables zero-shot transfer without modification to model weights. We provide the following prompts for the three target entity types: (i) “*disease or medical condition or diagnosis or disorder*”; (ii) “*symptom or clinical sign or finding or*

complaint”; (iii) “*medical procedure or surgery or test or treatment*”.

3.2 Sliding Window for Long Documents

Clinical documents in MultiClinAI average 1,055 tokens; 93% exceed 256 tokens and 76% exceed 512 tokens, while the model supports a maximum of 384 tokens. We implement a character-level sliding window with 800-character chunks and 150-character stride, extended to the nearest whitespace at boundaries. Predictions from overlapping windows are deduplicated by retaining the higher-confidence span.

3.3 Post-Processing Pipeline

Stage 1: Threshold Optimization. The model assigns a confidence score to each predicted span. We sweep thresholds from 0.2 to 0.8 in steps of 0.1 on a held-out 100-file English development sample, selecting the F1-maximizing threshold per entity type independently. This yields optimal thresholds of 0.8 for disease entities, 0.7 for symptom entities, and 0.4 for procedure entities. Higher thresholds for disease and symptom reflect the need to improve precision, which is substantially lower than recall in the zero-shot baseline (0.13–0.21 vs. 0.42–0.54).

Stage 2: Gazetteer Lookup. We maintain entity-specific lists compiled from DisTEMIST, SympTEMIST, and MedProcNER terminology resources. The gazetteer serves two functions: (i) label correction, where a predicted span whose surface form matches a known entity is reassigned to the correct label type; and (ii) gap-filling, where known entity surface forms not detected by the model are added via exact whole-word string matching. Based on offline evaluation, gazetteer augmentation is applied only to disease entities (+0.060 Strict F1), as it reduces precision for symptom and procedure entities due to ambiguous surface forms.

Stage 3: Span Boundary Correction. Predicted spans are trimmed of leading and trailing punctuation and whitespace, which the model frequently includes erroneously. Spans are additionally expanded leftward to capture common medical prefixes (*non-*, *post-*, *pre-*, *hyper-*, *hypo-*, *anti-*) when these immediately precede the detected span. Single-character spans are discarded as noise.

Stage 4: Negation Filtering. Entities appearing in negated contexts are removed using regular expressions that detect negation cues in the 50 characters preceding each span. Patterns include *no*, *not*, *denied*, *without*, *no evidence of*, and *no history of*. A set of generic section header terms (*medical history*, *chief complaint*, *physical examination*) are also excluded as they are not valid clinical entities.

4 Experimental Design

4.1 Dataset

The MultiClinAI dataset (Gallego-Donoso et al., 2026) (v1.2.1, Zenodo, March 2026) provides training data from four corpora: DisTEMIST, SympTEMIST, MedProcNER, and CardioCCC. Table 1 summarizes the statistics.

Language	Train Files	Annotations
English	3,774	79,316
Spanish	—	83,507
Czech	—	81,100
Dutch	—	80,853
Italian	—	81,482
Romanian	—	79,889
Swedish	—	80,190
All	—	566,337
EN test (batch 1)	16,657	—
All languages test	117,818	—

Table 1: MultiClinAI dataset statistics.

4.2 Baselines

XLM-RoBERTa (broken baseline): We initially trained an XLM-RoBERTa-base token classifier (Conneau et al., 2020) with MAX_LENGTH=256, which truncated 93% of documents, causing the model to predict every token as PROCEDURE (Strict F1 = 0.000). This illustrates a critical failure mode when document length is not accounted for in sequence labeling.

GLiNER-multilingual (zero-shot): urchade/gliner_multi-v2.1 evaluated offline achieved Strict F1 of 0.269, serving as our zero-shot general-domain baseline.

4.3 Evaluation Metrics

Following the shared task protocol, we report Strict Precision (P), Strict Recall (R), and Strict F1, which require exact match of both span boundaries and entity label. We additionally report CHR (character-level) F1, which awards partial credit for overlapping predictions. Offline evaluation

uses a 90/10 train/development split of the English training data; official metrics are computed by the shared task leaderboard.

4.4 Infrastructure

Development is performed on an Apple M1 Pro (16 GB) with MPS acceleration. Official test predictions are generated on a Tesla V100-32 GB on Bridges-2 (Pittsburgh Supercomputing Center) via a SLURM job array with one GPU per language. We use gliner v0.2.26, PyTorch v2.4.0, and transformers v4.44.0.

5 Results and Error Analysis

5.1 Official Leaderboard Results

Table 2 reports our official results on the English disease test set (batch 1).

System	P	R	F1	CHR F1
XLM-RoBERTa	0.000	0.000	0.000	—
GLiNER-multi	0.325	0.575	0.416	—
SuSh (ours)	0.554	0.486	0.518	0.613

Table 2: Official leaderboard results, English disease subtask.

5.2 Post-Processing Ablation

Table 3 shows offline development results across post-processing stages on the 100-file development sample.

Entity	Stage	P	R	F1
Disease	Baseline	0.134	0.493	0.211
	All combined	0.193	0.455	0.271
Symptom	Baseline	0.125	0.416	0.192
	All combined	0.153	0.313	0.205
Procedure	Baseline	0.213	0.542	0.306
	All combined	0.213	0.524	0.303
Overall	Baseline	0.157	0.489	0.238
	All combined	0.191	0.439	0.266

Table 3: Offline development results (100-file sample). Baseline = threshold 0.4, no post-processing. All combined = full four-stage pipeline.

The largest gain is observed for disease entities (+0.060 Strict F1), confirming the effectiveness of the disease-specific gazetteer. For symptom entities, the gain is smaller (+0.013) as the gazetteer is not applied to this type. For procedure entities, the change is negligible (−0.003), consistent with the observation that procedure spans tend to be

short, clean terms that benefit less from boundary correction.

5.3 Threshold Optimization Results

Table 4 presents the full threshold sweep for disease entities. At threshold 0.4 (the default), precision is only 0.325 while recall is 0.575, indicating substantial over-prediction. Raising the threshold to 0.8 improves precision to 0.554 at the cost of recall (0.486), yielding the best F1 of 0.518. This result directly matches our official leaderboard score, confirming that the development-set threshold generalizes to the test set.

Threshold	P	R	F1
0.2	0.245	0.742	0.369
0.3	0.287	0.659	0.400
0.4	0.325	0.575	0.416
0.5	0.354	0.484	0.409
0.6	0.394	0.373	0.383
0.7	0.397	0.221	0.284
0.8	0.554	0.486	0.518

Table 4: Threshold sweep for disease entities on the development set. Bold row is the F1-maximizing configuration.

5.4 Error Analysis

Why is Strict F1 lower than CHR F1? Our official results show Strict F1 of 0.5175 but CHR F1 of 0.6130, a gap of 0.095. This indicates the model frequently predicts the correct entity text but with slightly incorrect span boundaries. For example, the model predicts [107:125] ``tumour recurrence,`` (including the trailing comma) while the gold annotation is [107:124] ``tumour recurrence``. Strict matching penalizes this as both a false positive and a false negative, while CHR matching gives partial credit for the overlapping characters.

Precision-Recall Imbalance. We observe consistently high recall (0.42–0.54) but low precision (0.13–0.21) in the baseline. The zero-shot model over-predicts: it finds most real entities but also generates many false positives. Table 5 shows representative error cases.

Common False Positive Patterns. The most frequent false positives fall into three categories: (1) *section headers* such as “physical examination” and “chief complaint” are tagged as PROCEDURE or SYMPTOM entities despite being structural document elements rather than clinical findings;

Text span	Gold	Pred.	Error type
“physical exam.”	O	PROC.	Section header FP
“no fever”	O	SYMP.	Negation FP
“acute pain in the left...”	SYMP. (short)	SYMP. (long)	Boundary FP
“PKU”	DIS.	O	Abbreviation FN
“haemorrhagic pancreatic pseudocyst”	DIS.	O	Rare entity FN
“moderate”	O	DIS.	Context FP

Table 5: Representative error cases. FP = False Positive; FN = False Negative; DIS = DISEASE; SYMP = SYMPTOM; PROC = PROCEDURE.

(2) *negated entities* such as “no fever” are detected as positive entities because the model does not natively handle negation scope; and (3) *run-on spans* where a correctly identified entity start is followed by continued tagging across non-entity tokens, producing oversized spans such as “acute pain in the left hypochondrium” instead of “acute pain”.

Common False Negative Patterns. False negatives are dominated by two cases: (1) *abbreviations and acronyms* such as “PKU” (Phenylketonuria) or “COPD” are missed due to limited knowledge of clinical abbreviation conventions; and (2) *rare or compound disease names* such as “haemorrhagic pancreatic pseudocyst” or “moderate acalculous acute pancreatitis” are partially detected with incorrect span boundaries.

Gap Between Official and Offline F1. Our official Strict F1 (0.5175) substantially exceeds our offline development F1 (0.271). We attribute this primarily to the difference between training-domain development documents and the held-out test set, and to the 100-file development sample being a small and potentially unrepresentative subset of the full test distribution.

Limitations

The SuSh system is evaluated on the English disease subtask only, due to infrastructure constraints during the shared task period. Offline evaluation is restricted to a 100-file sample of the English training set. The zero-shot approach does not leverage available English training annotations, which we expect would yield substantial F1 improvements through fine-tuning. The gazetteer lists used are relatively small compared to full clinical ontologies such as UMLS or SNOMED-CT.

Conclusion

We presented SuSh, a zero-shot multilingual clinical NER system based on GLiNER-biomed-large-v1.0, achieving Strict F1 of 0.5175 on the Multi-ClinAI English disease subtask without any task-specific fine-tuning. Error analysis reveals that the primary limitations are span boundary imprecision (Strict-CHR gap of 0.095), negation handling, and abbreviation coverage. Future work will explore fine-tuning on the available English training data, integration of full UMLS ontology gazetteers, and ensemble approaches combining span-based and token-classification models to improve performance across all seven languages.

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